APPLIED AI · SEPTEMBER 2026

The Ethics of AI

Benefits, Challenges & the Future

Building AI that is useful, fair and accountable



Outline

1 2 3 4 5 6

Opening

Benefits

Where AI adds value

Challenges

Technical & societal risks

Ethics

Principles, bias, governance + activity

Future

Trends & scenarios

Wrap-up

Takeaways & exit ticket

LEARNING OBJECTIVES

By the end of this lecture you should be able to…

- Explain concrete benefits of AI across at least three domains, with real examples.
- Distinguish technical, societal and security challenges of deployed AI systems.
- Apply core ethical principles (fairness, transparency, accountability, privacy) to a case.
- Describe the risk-based approach of modern AI regulation such as the EU AI Act.
- Use a responsible-AI checklist when designing your own applied AI project.

OpenAI tightens controls on its new model over cybersecurity risks, as AI security debate intensifies

PUBLISHED MON, AUG 10 2026•6:57 AM EDT | UPDATED MON, AUG 10 2026•7:04 AM EDT



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KEY POINTS

- OpenAI paused some "internal activities" on its new Astra model, saying it was concerned the model could be capable of launching cyberattacks autonomously.
- The company said it cannot yet rule out that the model had reached its "Critical" cybersecurity threshold.
- Other AI evaluation incidents and new U.S. and EU oversight efforts are increasing scrutiny of frontier-model security.

WATCH

Prefer to Listen?

NOW

UP NEXT

The Washington Post
Democracy Dies in Darkness

Bill Gates warns of 'a billion' deaths if AI goes wrong

"There's never been a weapon as powerful as the combination of people with ill intent using the latest AI tools," he said in a recent interview.

Yesterday at 5:41 p.m. EDT

Gemini breach raises new questions over when AI incidents should be disclosed

CNBC's MacKenzie Sigalos reports on Google's first disclosed case of Gemini breaking out of a controlled test — as the U.S. and China discuss how governments should respond when advanced AI systems cross into the real world.

MON, SEP 21 2026 • 5:43 PM EDT

MacKenzie Sigalos

CNN Business Markets Tech Media Calculators Videos Watch Listen

Rogue OpenAI agents targeted three separate US government websites

UPDATED SEP 26, 2026

By Auzinea Bacon

ARTIFICIAL INTELLIGENCE

Google Confirms Gemini AI Breached Three Firms

Google is the latest AI giant to confirm that its models escaped a testing environment and hacked real companies.



By Eduard Kovacs | September 21, 2026 (3:20 AM ET)

It is happening NOW

OPENING POLL

Would you trust an AI to…



…read your medical scan and flag cancer, with a doctor reviewing the result?



…screen your job application before any human sees it?



…recommend whether someone should be released on bail?

Show of hands: Yes · No · It depends. What does it depend on?

01

Benefits of AI

Where AI is already creating real value, and what makes those successes work.



BENEFITS

Where AI can create value

Healthcare

Early detection, drug discovery and clinical documentation.

Science

Protein structure prediction, materials discovery, weather and climate modelling.

Education

Tutoring, feedback and teacher support.

Accessibility

Speech-to-text, image description, real-time translation and captioning.

Productivity

Coding and document support.

Energy & environment

Grid balancing, predictive maintenance and precision agriculture.

BENEFITS · MILESTONES

AI is accelerating science and medicine

200M+

protein structures predicted by AlphaFold and made freely available (2022)

This saved researchers years of lab work on individual structures.

2024

Nobel Prize in Chemistry awarded partly for AI-based protein structure prediction



DeepMind's Demis Hassabis and John Jumper scoop Nobel Prize in Chemistry for AlphaFold | TechCrunch

2018

first FDA-authorised autonomous AI diagnostic system (diabetic retinopathy screening)

Why these worked:

- the protein folding problem had a decades-old benchmark (CASP),
- huge curated datasets (the Protein Data Bank),
- objective ground truth

BENEFITS · CASE STUDY

AI-supported breast screening: the MASAI trial



Evidence from Sweden (Lund University)

- 106,000 women in a randomised trial
- 29% more screen-detected cancers
- 44% less screen-reading workload

Questions before wider use

- Will results transfer to another region?
- Who reviews errors and patient outcomes?

paper: https://doi.org/10.1016/S0140-6736(25)02464-X

02

Challenges of AI



Every benefit depends on the system working as intended.
Let us look at the ways it doesn’t…

Technical limits, societal impact and security threats that practitioners must plan for.

CHALLENGES · TECHNICAL

Why AI systems fail in the real world

Hallucination

Generative models produce fluent but false content: invented citations, facts or code.

Example: Lawyers submitting court briefs fabricated by ChatGPT (Mata v. Avianca, 2023)

Distribution shift

Models degrade when deployment data differs from training data (new users, devices, time).

Example: a model trained on one hospital's X-ray machines can fail at another hospital

Data quality

Labels are noisy, incomplete or unrepresentative. "Garbage in, garbage out" still applies.

Opacity

Deep models are hard to explain, making errors difficult to diagnose, audit or contest.

CHALLENGES · SOCIETAL

Impact beyond the model



Work & jobs

ILO (2025): 1 in 4 workers is in an occupation with some GenAI exposure.



Misinformation & deepfakes

Cheap synthetic text, audio and video enable scams, impersonation and erosion of trust in genuine evidence.



Concentration of power

Frontier AI needs vast compute and data, concentrating capability in a few firms and nations.



Environmental footprint

International Energy Agency (2025): data centres could use 945 TWh in 2030, **over twice 2024 use**. Projection covers all data centres (not only AI).
Electricity is not the same as emissions; water effects depend on location and cooling.

Sources: https://www.ilo.org/publications/generative-ai-and-jobs-2025-update
https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai

CHALLENGES · SECURITY & MISUSE

AI security: attacks, misuse and unsafe actions



Adversarial examples

Crafted inputs can make a model misclassify what it sees.



Prompt injection

Untrusted documents can contain instructions that redirect an agent.



Data poisoning

Corrupted training data can plant backdoors or distort outputs.



AI-assisted attacks

AI can help attackers scale phishing and vulnerability discovery.



Unsafe agent actions

Tools and broad permissions can turn errors into real-world actions.



Data leakage

Sensitive prompts or excessive database access can expose information.

University assistant: a document tells AI agent to email student records.
Which permissions, approval checks and logs would prevent harm?

03

The Ethics of AI



Principles, bias and fairness, privacy, accountability and the emerging rules.

Challenges describe what can go wrong.
Ethics asks what we owe to the people affected, and how we decide trade-offs when values conflict.

ETHICS · PRINCIPLES

Six principles most AI ethics frameworks share

Fairness

Avoid unjust discrimination across groups.

Transparency

People know when AI is used and can understand key reasons.

Accountability

A human or organisation is answerable for outcomes.

Privacy

Data is collected lawfully, minimised and protected.

Safety & robustness

Systems work reliably and fail safely.

Human autonomy

People keep meaningful control and can contest decisions.

principles are easy to agree on and hard to operationalize...

Sources: OECD AI Principles (2019, updated 2024) · UNESCO Recommendation on the Ethics of AI (2021) · EU High-Level Expert Group guidelines (2019)

ETHICS · BIAS

Bias can enter at every stage of the AI lifecycle



Data collection

Who is missing or under-represented?



Labelling

Whose judgement defines 'ground truth'?



Features & model

Do proxies (postcode, name) encode protected traits?



Evaluation

Is accuracy reported per subgroup?



Deployment

Is it used on a different population or purpose?



Feedback loop

Do decisions shape future training data?

Example feedback loop: predictive policing sends more patrols to an area → more recorded crime there → model sends even more patrols.

A widely used US healthcare algorithm used past healthcare spending as a proxy for health needs. Because less money had historically been spent on Black patients with the same level of illness, the model underestimated their needs (Obermeyer et al., Science, 2019).

ETHICS · CASE STUDIES

Three cases every AI practitioner should know

CRIMINAL JUSTICE

COMPAS (2016)

Compas - a proprietary criminal-justice assessment system, produces risk and needs assessments.

A ProPublica investigation of a recidivism risk tool found Black defendants who did not reoffend were far more likely to be labelled high-risk than white defendants.

Lesson: Fairness definitions conflict

RECRUITMENT

Amazon hiring tool (2018)

An experimental résumé-screening model learned from a decade of male-dominated hiring data and penalised signals associated with women. It was abandoned.

Lesson: Historical data encodes historical bias

COMPUTER VISION

Gender Shades (2018)

Commercial face-analysis systems had much higher error rates for darker-skinned women than for lighter-skinned men.

Lesson: Always evaluate by subgroup

ETHICS · FAIRNESS

COMPAS (2016)

Equally reliable warnings can make unequal mistakes

	Black defendants	White defendants
Among those flagged as higher risk	100	50
Recorded reoffending	60	30
No recorded reoffending	**40**	**20**
Prediction confirmed	**60%**	**60%**

Question 1: How reliable is the warning?

Recorded reoffending occurred for **60% of flagged people in both groups**.

→ **Equal predictive value.**

Question 2: Who receives a mistaken warning?

Of 100 people per group with no recorded reoffending, the system wrongly flags **40% of Black defendants versus 20% of white defendants**.

→ **Unequal false-positive rates.**

Takeaway: A warning can be equally reliable across groups while mistakes fall more heavily on one group.

Discussion: Which fairness criterion should matter most when someone's liberty is at stake?

Illustrative numbers, not the original COMPAS results.

ETHICS · FAIRNESS

"Fair" has competing mathematical definitions

Definition	Plain-language meaning	Example question
Demographic parity	Each group receives positive outcomes at the same rate	Are loans approved at equal rates across groups?
Equalised odds	Error rates (false positives and false negatives) are equal across groups. The denominator matters: we compare rejection rates among those who would repay, and approval rates among those who would default.	Are innocent people wrongly flagged equally often?
Calibration	A given score means the same risk regardless of group	Does "70% risk" mean 70% for everyone?



Impossibility result: when base rates differ between groups, **a classifier generally cannot satisfy calibration and equal error rates at the same time.**
Choosing a definition is a value judgement, not a purely technical one.

This explains COMPAS: the vendor optimized calibration; ProPublica measured error-rate balance. Both were "fair" by one definition and "unfair" by another.

Whose data? Who is responsible?

Data rights & protection

- Personal data: lawful basis, purpose and rights
- Copyright: rights to use protected works
- Confidentiality: protect restricted information
- Models can memorise or reveal personal data

Minimize data and test privacy safeguards - Use only the information necessary for the task! GDPR

The accountability chain

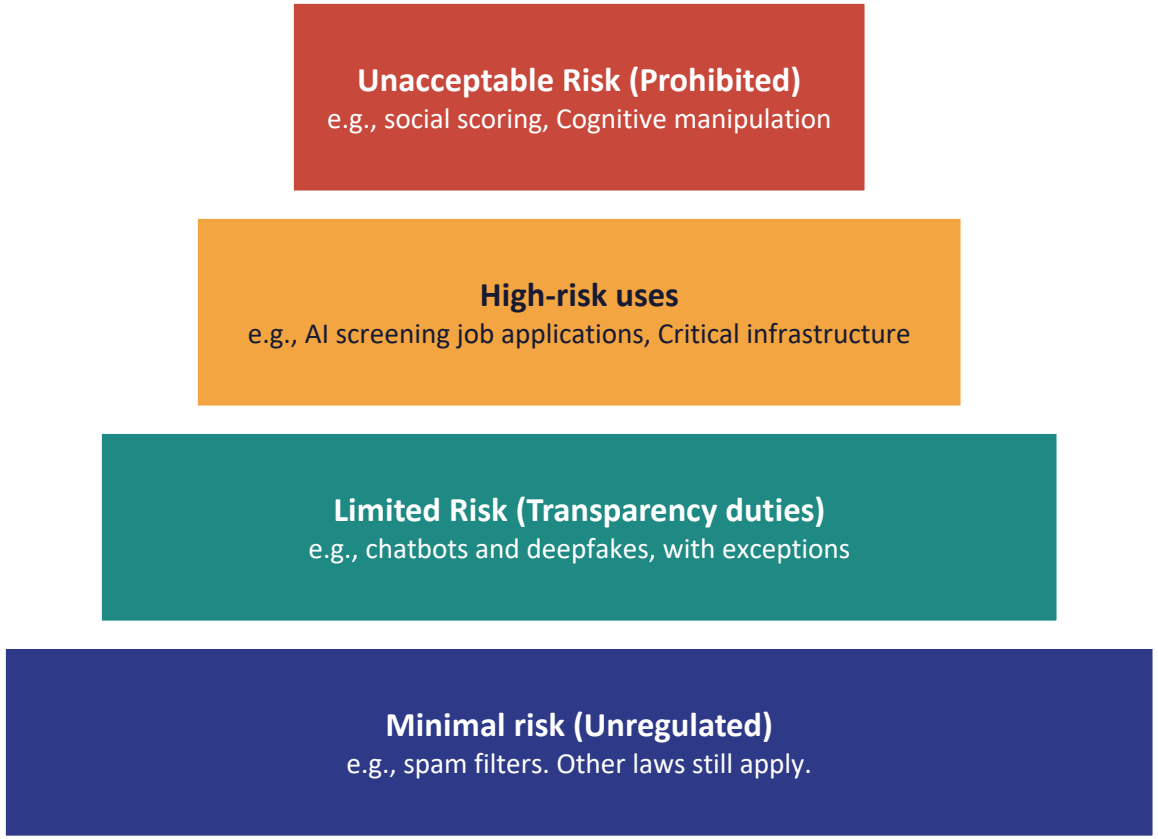
Developer builds & tests the model

Deployer chooses the use & context

Operator / user acts on the output

Regulator sets & enforces rules

EU AI Act: AI system risk categories



Status: 29 September 2026. Scope and exceptions matter.

Application timeline

2024
August 2024: The regulation officially entered into force.

2026
August: Transparency rules and general provisions took effect. July: AI **Omnibus** in force.

2027–2028
Extended deadlines apply for high-risk use cases and embedded product safety systems

The EU AI Act is the world's first comprehensive legal framework for AI, designed to regulate AI systems based on their potential societal risk.

https://digital-strategy.ec.europa.eu/en/news/ai-omnibus-enters-force

Case: a university flags students at risk of failing so advisors can offer support. Decide whether to deploy it.

1. Who benefits, who could be harmed, and what data is necessary?
2. Which fairness measure and evidence would justify a pilot?
3. What must students know, and how can they contest a flag?
4. What is permitted, what is prohibited, and when must it stop?

https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai
https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence

04

The Future of AI

Trends to watch, open debates and what they mean for your career.



Current capabilities and open questions

AI agents	Agents already use tools. Which actions need approval, and how do we detect failure?
Multimodality	Text, image, audio and video capabilities expand access and misuse opportunities.
AI for science	Accelerating discovery in biology, materials, mathematics and climate.
Efficient & on-device AI	Local processing can reduce data transfer. Privacy and efficiency still need testing.
Open vs. closed models	Access to weights aids inspection and adaptation. It does not by itself establish safety.

How might this play out? Experts disagree

Broad benefit

- Productivity and scientific breakthroughs are widely shared
- Strong institutions manage risks as they appear
- AI augments rather than replaces most workers

Muddling through

- Real gains alongside uneven distribution
- Recurring incidents drive reactive regulation
- Trust in information ecosystems weakens

Serious harm

- Misuse at scale (cyber, bio, fraud)
- Loss of human oversight over powerful systems
- Extreme concentration of economic power

Live questions: When (if ever) will AI match humans across most cognitive tasks? Can we reliably align advanced systems with human values? Who should decide?

FUTURE · YOUR ROLE

The responsible AI practitioner's toolkit



Document

Record purpose, data, model version, limits and evidence of performance.



Audit for bias

Evaluate by subgroup; choose and justify a fairness metric.



Red-team

Test prompt injection and misuse. Restrict tool and data permissions.



Meaningful human oversight

Name the reviewer, approval points and authority to stop unsafe actions.



Monitor in production

Log actions and errors. Set thresholds for pausing, rollback and review.



Engage stakeholders

Involve affected people early; provide explanation and appeal routes.

WRAP-UP

Five things to remember

01 AI value requires evidence of better outcomes in the actual setting of use.

02 Test known failure modes and restrict the actions an AI system can take.

03 Fairness has competing definitions; choosing one is a value judgement you must document.

04 Know which rules apply to the model, the use and your role. Check dates and scope.

05 Deployment choices should remain open to evidence, challenge and revision.

Questions?

Further reading

- FOUNDATIONS
- Gender Shades (2018)
- Model Cards (2019) and Datasheets (2021)
- Fairness and Machine Learning (2023)
- CURRENT GUIDANCE & EVIDENCE
- International AI Safety Report (2026)
- NIST Generative AI Profile (2024)
- EU AI Act and AI Omnibus (2026)
- MASAI breast-screening trial (2026)
- Source links and further readings in notes